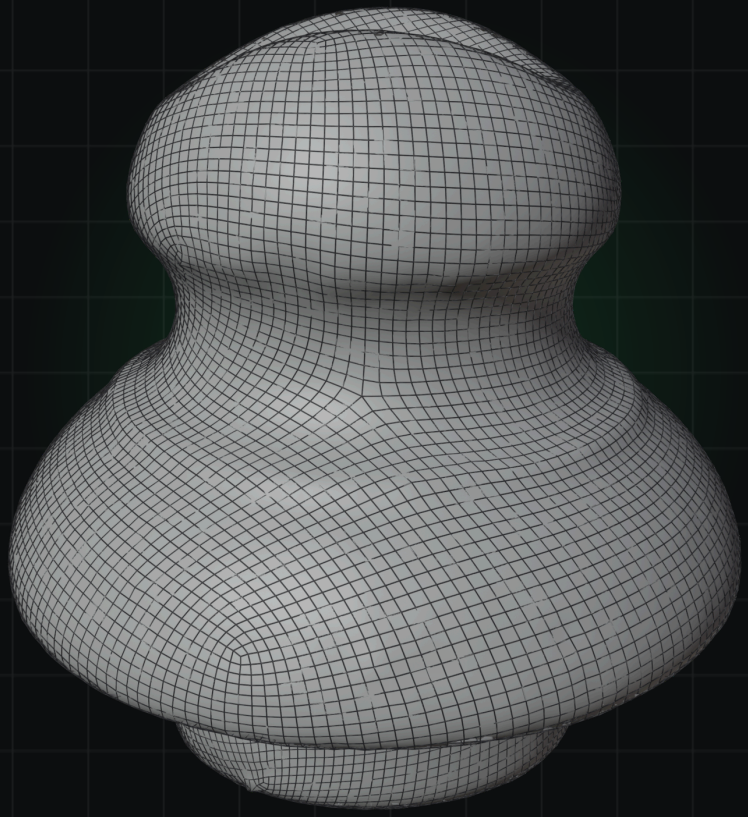
Raw scan in. Clean mesh out.

Clean, repair, reduce and quad-remesh dense 3D scans without leaving Blender. Every run works on a copy, so your original scan is never touched.



14,534 QUADS · WATERTIGHT

3.19M → 300k
triangles in one run,
61 s on a 2-core machine

75 shells
hidden interior fragments
found and removed

0 open edges
watertight result,
ready for slicer or Boolean

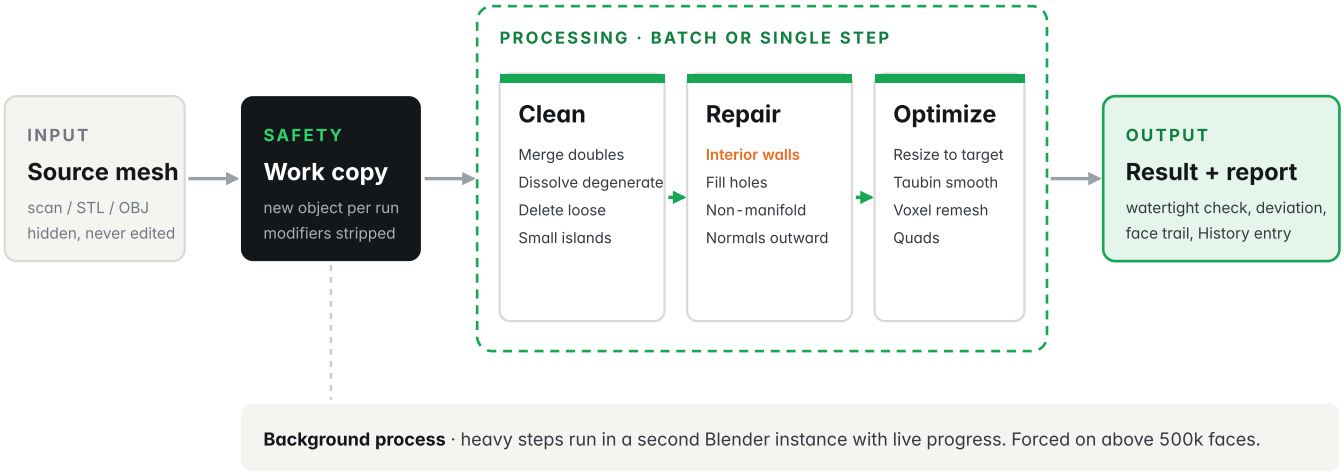
12 steps
run as one batch
or one at a time

Blender 4.2 and newer · No extra Python packages · GPL-3.0-or-later Test part: insulator_001.stl, raw 3D scan, 3.19M triangles

01 HOW IT WORKS

One panel. One pass. Your source stays safe.

Select a mesh, tick the steps you need and press **Run all selected**. MeshPharmacy duplicates the object, hides the original and runs every step on the duplicate. Want to test a single operation first? Press ▶ next to it instead.



▶ Run all selected Batch

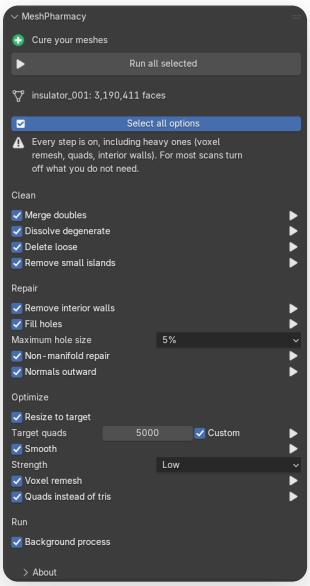
All ticked steps run on **one** copy, in a fixed safe order. Use it when you know what the scan needs and want a finished mesh in one click.

Result: `insulator_001_300k_tris_outer`

▶ Single step

Every ▶ button runs only that step on **its own** copy. Run several and compare them side by side before you decide what to keep.

Result: `insulator_001_interior` , `insulator_001_smooth`



Sidebar › MeshPharmacy.
Press N in the 3D Viewport.

Batch order

The order is fixed so steps do not undo each other. Unticked steps are skipped.

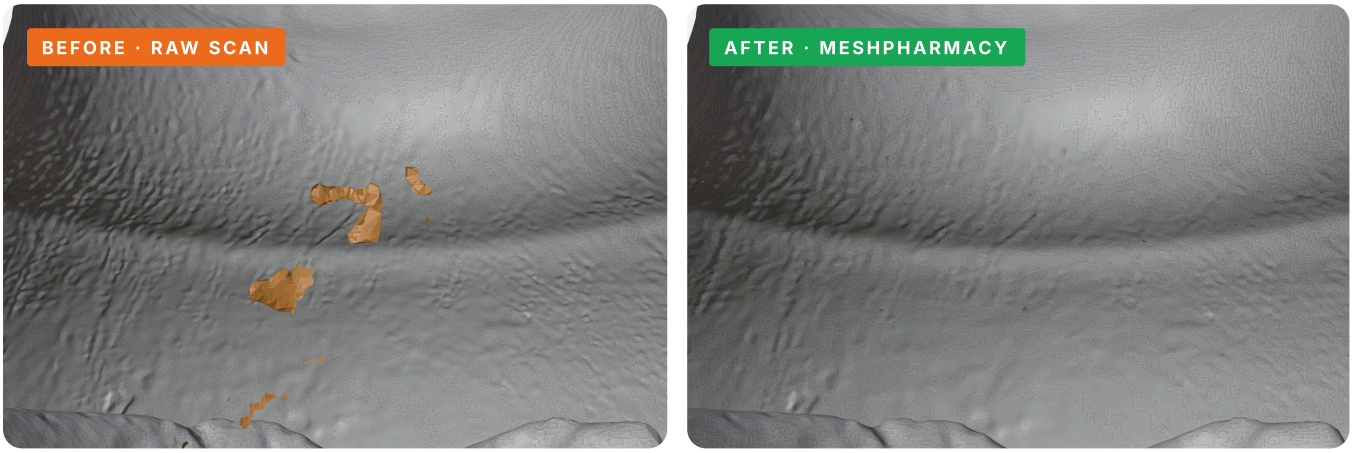
01	Work copy · source hidden
02	Merge, degenerate, loose
03	Remove small islands
04	Remove interior walls
05	Voxel remesh
06	Resize to target
07	Non-manifold repair, fill holes
08	Normals outward
09	Taubin smooth
10	Quads · QuadriFlow
11	Watertight + deviation check
12	Report to History

Always a copy. The source object is hidden, not changed. Every run adds a new, clearly named object to the same collection. Ctrl+Z works as usual, and **Restore visibility** brings back exactly the objects a run hid.

02 CLEAN & REMOVE INTERIOR WALLS

The junk you can't see is the junk that breaks prints.

Scanners often record surfaces through openings or from overlapping passes. They end up as loose shells **inside** the part. In the viewport the model looks fine. In the slicer you get phantom walls, wrong infill and failed Booleans.



Section view through insulator_001, upper part. The scan was cut in half to show the inside of the neck. The orange shells are separate pieces of geometry floating inside the body. Nothing else on the surface was changed.

78	75	1	26 s
separate shells in the raw scan	of them sit inside the body (7,347 faces)	clean outer shell after the run	interior check, 3.19M faces

Clean

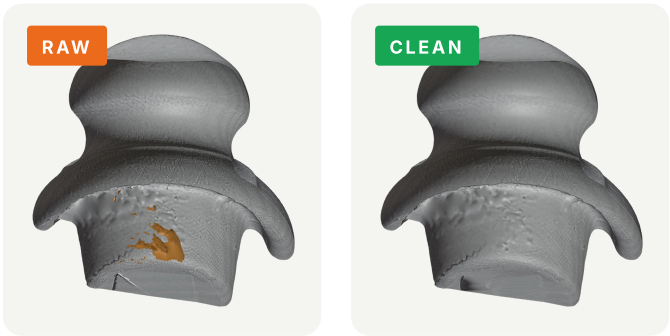
Welds duplicate vertices (distance scaled to the object size), dissolves zero-area faces and deletes loose vertices and edges. **Remove small islands** deletes shells smaller than 5% of the main body.

Remove interior walls

Tests every remaining shell against the main body with ray casting on all three axes. If a shell is inside, it is removed **whatever its size**. That covers big inner surfaces the island filter keeps.

Built-in safety limits

If island removal would delete more than 10% of the faces, or interior removal more than 30%, the step is skipped and the report tells you why. A fragmented scan is never silently gutted.



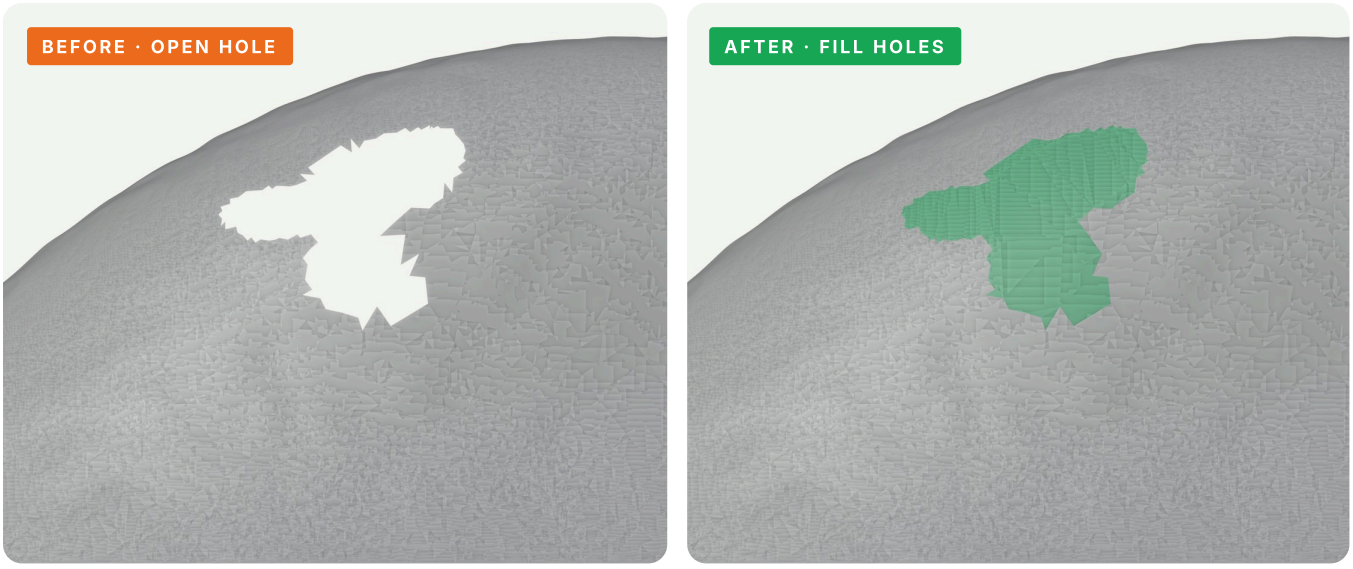
Full section. The close-up above is taken from the inside of the neck. The part is hollow, as most scans are. Only the outer skin is kept.

> **Tip:** Remove interior walls is off by default because it adds time on huge scans. Turn it on for anything going to a slicer, a Boolean or a volume calculation.

03 REPAIR & VERIFY

Watertight, checked, and reported.

Repair closes the surface and fixes broken topology. Then MeshPharmacy checks the result itself, so you don't need the 3D-Print Toolbox to know whether it worked.



Fill holes on the top of insulator_001. An irregular hole (97 open edges) was cut into the scan on purpose. One Fill holes step closed it with 95 new faces, shown in green. Result: 0 open edges, closed volume.

CHECK	RAW SCAN	RESULT
Triangles	3,190,411	299,994
Separate shells	78	1
Open boundary edges	1,195	0
Open boundary loops	58	0
Pinched vertices	1	0
Multi-face edges	0	0
Watertight	No	Yes

WATERTIGHT: YES

```
Result    insulator_001_300k_tris_outer
Faces     3190411 → 299994 triangles
Size      95.35 × 94.12 × 99.92 (unchanged)
Deviation mean 0.0017, max 0.0088
Islands   77 removed
Holes     filled, left open: 0
Topology  open=0 multi=0 pinched=0
Time      61.0 s

Face trail
3190411 › clean 3190363 › islands 3183008 ›
reduce 300000 › repair 299994
```

Real History report from the full batch run on insulator_001 (lightly trimmed). Deviation is sampled result-to-source distance in scene units. The **face trail** shows which step removed how many faces.

What Repair does

- **Fill holes** up to a size you choose: 0.5%, 2%, 5% of the object size, or all.
- **Non-manifold repair** removes faces on edges shared by 3+ faces and around bow-tie vertices. It stops before deleting more than 3% of the mesh.
- **Normals outward** makes every face point out, as slicers expect.

04 REDUCE & SMOOTH

Fewer triangles. Same part.

A raw scan carries millions of triangles you don't need. **Resize to target** collapses the flat areas first and keeps triangles where the shape changes, so a 160 MB file becomes 15 MB and the part still measures the same.



3,183,054 159 MB STL



900,000 45 MB STL



300,000 15 MB STL



99,998 5 MB STL

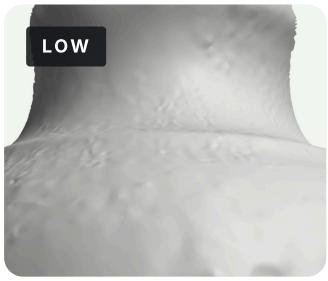
TARGET PRESET	TRIANGLES	STL SIZE	MEAN DEVIATION	MAX DEVIATION	VOLUME CHANGE	TIME
Raw (clean shell)	3,183,054	159 MB	–	–	–	–
900k	900,000	45 MB	0.0005	0.0025	−0.0002 %	20 s
300k · default	**300,000**	**15 MB**	0.0017	0.011	−0.0014 %	24 s
100k	99,998	5 MB	0.0047	0.049	−0.0052 %	25 s

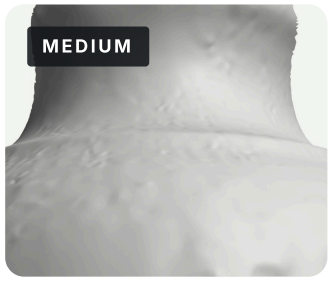
Deviation: sampled distance to the raw surface in scene units, part about 100 units tall. Below the target, the step subdivides instead.

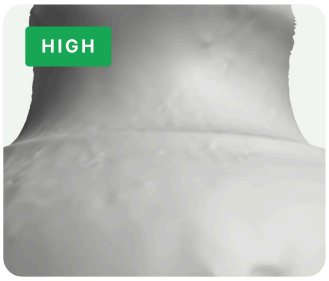
Smooth in three strengths

Taubin filter · passes shown for ~300k faces, scaled to density









Neck of the 300k result, real scan surface, nothing added. Scanner grain goes first; at High the larger dents soften too, while the ring stays sharp.

Low `λ 0.5 · 5 passes`

Takes the grain off the surface. Use it on scans that are already clean or when you need every detail.

Medium `λ 0.5 · 14 passes`

The everyday setting. Removes scanner noise and keeps the shape crisp.

High `λ 0.6 · 32 passes`

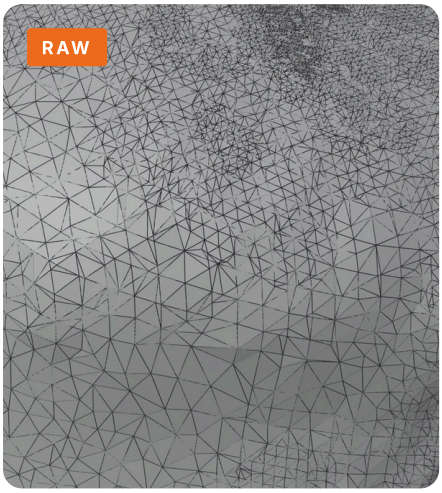
For noisy scans and parts that will be painted or printed. Still no visible shrinkage.

No shrinking. Plain smoothing pulls the surface inward. Taubin pushes it back out after every pass. Volume change in this test: Low +0.007 %, Medium +0.013 %, High +0.032 %. Passes adapt to mesh density, and above 2 % the report warns you.

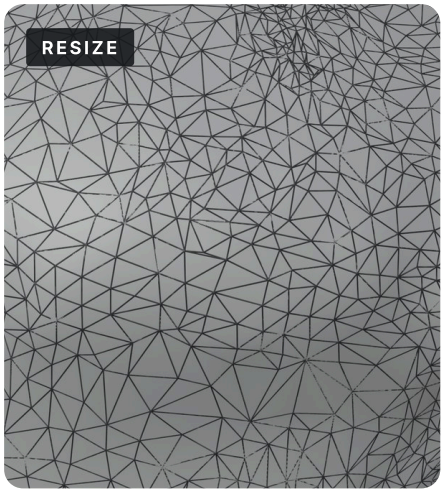
05 OPTIMIZE

From scanner noise to a mesh you can actually work with.

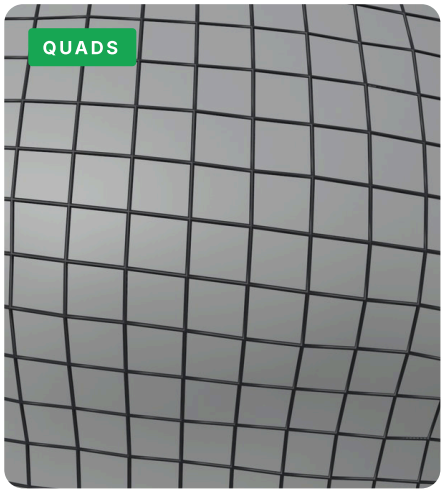
Set a target and MeshPharmacy decimates or subdivides to reach it. Then you can smooth out scan noise without shrinking the part, or rebuild the surface as clean quads.



Raw scan 3,190,411 tris



Resize to target 299,994 tris



Quads instead of tris 14,534 quads

Same spot on the part at the same zoom. Quad result: watertight, mean deviation 0.018 and max 0.22 scene units on a 100-unit part.

Resize to target	Presets 100k · 300k · 600k · 900k · 2M, or a custom count. Decimates when above, subdivides when below (capped at 2M for memory safety).
Smooth	Taubin filter: Low, Medium or High. Removes scanner noise without the shrinkage of plain Laplacian smoothing. Warns if volume changes more than 2%.
Voxel remesh	Rebuilds a uniform surface from voxels at 0.2% of the object size. Slow, and drops UVs and vertex colors.
Quads	Blender's built-in QuadriFlow, aimed at your target count. If it fails, triangle pairs are joined instead and the report says so.

Recommended first run

- Keep Clean and Repair on, add **Remove interior walls**.
- Resize to **300k**, leave Smooth, Voxel and Quads off.
- Check the report. If needed, run Smooth or Quads as **single steps** on the result and compare.

> Target counts are approximate: cleanup, repair and QuadriFlow all change the final number slightly.

Install

Edit › Preferences › Add-ons › Install from Disk › `mesh_pharmacy.zip`. Open with N › MeshPharmacy.

Requirements

Blender 4.2 or newer. No extra Python packages, only Blender's bundled modules.

Large scans

Multi-million-face meshes need RAM and time. Both Blender processes share your machine, so start with Resize only.

License

GPL-3.0-or-later. Full source included. © 2026 Mirosław Pałucki.